Accelerated Math II/III Names: ______________________
HW #1 – Quadratic Functions Period: ____________________

Complete #3, 6, 9-15, 18-27 (mult of 3), 37. Do not use your graphing calculator, must show all work and find the vertex and axis of symmetry algebraically. Graph on graph paper.

Find the equation of the axis of symmetry and the coordinates of the vertex of the graph of each function. Find the domain and range

1. $y = x^2 - 10x + 2$

2. $y = x^2 + 12x - 9$

3. $y = -x^2 + 2x + 1$

4. $f(x) = 3x^2 + 18x + 9$

5. $y = 3x^2 + 3$

6. $f(x) = 16x - 4x^2$

7. $y = 0.5x^2 + 4x - 2$

8. $y = -4x^2 + 24x + 6$

9. $y = -1.5x^2 + 6x$

Match each graph with its function.

A. $y = -x^2 - 3x$

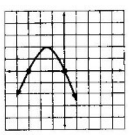
B. $y = x^2 - 3x$

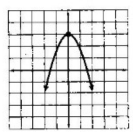
C. $y = x^2 + 3x$

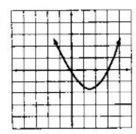
D. $y = -x^2 + 3$

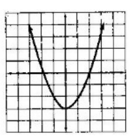
E. $y = x^2 - 3$

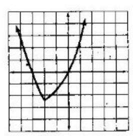
F. $y = -x^2 + 3x$

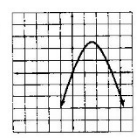
10. 

11. 

12. 

13. 

14. 

15. 

Graph each function. Label the axis of symmetry and the vertex. Plot a minimum of 5 points.

16. $y = x^2 - 6x + 4$

17. $f(x) = x^2 + 4x - 1$

18. $y = x^2 + 10x + 14$

19. $y = x^2 + 2x + 1$

20. $y = -x^2 - 4x + 4$

21. $f(x) = -4x^2 + 24x + 13$

22. $f(x) = -2x^2 - 8x + 5$

23. $y = 4x^2 - 16x + 10$

24. $y = -x^2 + 6x + 5$

25. $y = 4x^2 + 8x$

26. $f(x) = -3x^2 + 6$

27. $y = 6x^2 + 48x + 98$

37. You and a friend are hiking in the mountains. You want to climb to a ledge that is 20 ft above you. The height of the grappling hook you throw is given by the function $h = -16t^2 - 32t + 5$. What is the maximum height of the grappling hook? Can you throw it high enough to reach the ledge?